

**General Description of
Electrical System**

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1 SCOPE

The scope of this document is the General description of the Electrical system and definition of minimum technical requirements for the design, supply, installation and commissioning of the various electrical systems and equipment for the Power plant supplying power to the SEZ NIPT

The power plant will be located within the IPC, which will be a world class petrochemical production facility will be located close to Karabatan, to the north-east of the regional capital of Atyrau.

2 REGULATIONS

The following documents, including applicable addenda, constitute a part of this specification as described herein. The referenced edition of these documents shall be that which is in effect at the issue date of this specification. Detailed design, and subsequent equipment supplied, shall comply with the latest edition of the standards listed below:

2.1 Kazakhstan Standards and Codes

- Electrical equipment and materials shall be in accordance with KAZAKH ORGANIZATION'S PUBLISHED STANDARD requirements, unless specified otherwise.
- PUE RK 2003 Electrical Installation Codes
- GOST21.607-82 (ed.2003) SPDS. Electric lighting of the territory of industrial enterprises. Working drawings.
- GOST21.608-84 (ed.2003) SPDS. Internal electric lighting. Working drawings.
- GOST21.613-88 (ed.2003) SPDS. Electrical power equipment. Working drawings.
- SN RK 3 02-19-2001 Construction of electricity supply facilities in seismic areas
- SNiP II-58-75* Thermal power stations
- RD 34.03.355-90 Manual on the provision of explosion safety during designing and maintenance of power gas turbine installations
- SNiP RK 4.04-10-2002 Electrical devices
- SNiP RK 2.04-05-2002* Natural and artificial lighting

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- ONTP 5-78 Technological standards of high voltage substation design with voltage 35-750kV
- SN RK 2.04-29-2005 Manual on the lightning protection of the buildings and facilities
- GOST14254-96 (MEK 529-1989) Ingress protection of electrical equipment "IP" "International Protection" ; "Ingress Protection"

2.2 IEC Standards

Electrical equipment and materials shall comply with the latest edition of all relevant IEC publications, supplemented by narrative national specifications, (or their functional equivalents).

2.3 Project Specifications (except as modified here in this specification)

Contractor to develop.

2.4 Terms and Abbreviations

AOLTC	Automatic, On-Load, Tap Charger (Transformers)
ATS	Automatic Transfer Switch / System
BRA	Building Risk Assessment
CB	Circuit Breaker
CCB	Centralized Control Building
DC	Direct Current Uninterruptible Power Supply
DCS	Distributed Control System
DG	Diesel Generator
EMCC	Emergency Motor Control Center
EMI	Electromagnetic Interference
EPS	Emergency Power System
ETAP	Electrical Transient Analyzer Program
EM	Emergency
F&G	Fire and Gas
GTG	Gas Turbine Generator
HV	High Voltage

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HVAC	Heating, Ventilation and Air Conditioning
IPC	Integrated Petrochemical Complex
JB	Junction Box
LAN	Local Area Network
LSS	Load Shedding System
LV	Low Voltage
MCC	Motor Control Centre
MOV	Motor Operated Valve
MV	Medium Voltage
NER	Neutral Earthing Resistor
O/H	Overhead
PCS	Process Control System
PDCS	Power Distribution & Control System
PMCS	Power Monitoring & Control System
PMS	Power Management System
RTD	Resistance Temperature Detector
SCADA	Supervisory Control and Data Acquisition
SEZ NIPT	Free Economic Zone National Industrial Petrochemical Park
SRR	Satellite Rack Room
STG	Steam Turbine Generator
SWG	Switchgear
UPS	Alternate Current Uninterruptible Power Supply
VSD	Variable Speed Drive
Shall and Must	indicate a mandatory requirement.

3 SITE CONDITIONS

The site is located in the Republic of Kazakhstan by the northern coast of the Caspian Sea within 50 km of the city of Atyrau. The territory of the Atyrau region represents a low flat country or a slightly elevated flat country, located in a semi-desert and desert zone. Being in

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North-East coast of the Caspian Sea, wind borne dust and sand storms over the flat terrain are a serious problem. Corrosion in the area is high and extremes of temperature over the year require special solutions for construction, operations and maintenance activities.

The design parameters are given in: 01000-EM-SPC-003 General Site Information: Meteorological, Geological And Utilities Data. The Power Plant shall be able to generate steam and electric power to its design capacity, in the summer and winter conditions specified hereafter, as well as in any intermediate climatic condition.

Substation design temp range 15 / 30°C

Substation design humidity 75% maximum

Power Plant Contractor shall allow for any possible impact of extreme low temperatures on the design and operation of the power plant.

4 AREA CLASSIFICATION

To determine the extent of the precautions required to prevent the ignition of potentially explosives atmospheres, areas of plants shall be classified into hazardous zones (if any) as per IEC 60079.

The area classification shall be based on latest edition of Part 15 of the Institute of Petroleum model code of safe practice. Grouping of various hazardous gases and vapours shall be as per IEC 60079. Electrical equipment selection and installation for classified area shall be in accordance with IEC 60079. The Area Classification drawings shall be prepared from the project plot plans and the Process Schedule of Hazardous Materials to determine potential sources of hazard. The Schedule of Hazardous Materials shall be developed by the process groups to identify the potential sources of release and the quantities and type of medium which might be released. The electrical group shall take this as an input to the Area Classification process, and develop the documents for submission to the review meetings. Areas of plant not classified under the above procedures shall be designated as being non-hazardous.

When a classified area extends over an uncontrolled roadway, it shall be reviewed, and if the risk is judged unacceptable by Company, then steps shall be taken to reduce the risk.

The degree of protection for electrical and instrument enclosures shall be a minimum of Ex-d IP65 for all outdoor equipment. All outdoor equipment shall be designed to operate in a severe corrosive environment.

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All equipment located in hazardous areas shall be labeled or listed or certified by a recognized Certification Agency. Any electrical equipment and materials, intended for use in explosion hazardous area must be able to provide a hazardous area certification approval by the Kazakhstan Authorities.

5 STANDARDIZATION

The electrical system design, selection of equipment and proposed installation shall ensure the highest achievable levels of plant reliability and availability within the confines of the relevant codes, standards and project design basis.

All electrical equipment across the complete project shall be standardized, within practical limits. Where this requirement cannot be met, then full compatibility between the particular equipment supplied and other equipment on the Project shall be guaranteed by Power Plant Contractor. Particular attention should be given to elements of electronic networks across the sites (such as PMS, PMCS, LSS, etc.), where full compatibility of functions, data handling and storage, and connectivity is vital to ensure a fully functioning system.

It is anticipated that to achieve the above aims, equipment will be sourced from a minimum number of Subcontractors, and, where necessary, Power Plant Contractor will guarantee compatibility of similar equipment from different Subcontractors.

5.1 Materials

The extent of standardization shall ensure that similar items of equipment are selected from a single Supplier, and are of the same Manufacturer's generic model/type and standard rating. The expectation is that standardization will result in

- a reduced requirement for spare parts holdings
- an easing of equipment and systems familiarization for installation, commissioning and maintenance.

Power Plant Contractor shall consider a methodical phased approach to equipment replacement and installation. This approach shall ensure continuity of supply is maintained and that commissioning time is minimized.

5.2 Installation

It is anticipated that the Power Plant Contractor will develop typical installation details for each multiple system installation, such as sockets, lights, junction boxes, earth bars, area lighting

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poles, etc., and submit these for Company approval. These will be subject to customization for certain installations.

Equipment shall be located with sufficient spacing to permit future extensions to enclosures, access to all components for normal maintenance work or complete removal, if required.

The layout of the facilities shall permit adequate maintenance access to all components of the installed systems. No permanent maintenance facilities should be required. It should be possible to perform all major maintenance activities using the mobile equipment provided.

All special tools required by Company for testing and future maintenance activities shall be provided with the main electrical equipment. These may be grouped to a specific location – such as a single set per substation.

6 ENERGY CONSERVATION

It is important that the plant running energy costs are kept to a minimum, so Power Plant Contractor shall demonstrate to Company its widespread use of high efficiency equipment, particularly motors and lighting fittings. In the Power Plant, the overall efficiency of the whole plant is essential to the choice of equipment manufacturers as well as a good engineering service.

It is anticipated that the overall power factor at the IPC complex shall not be less than 0.9, and Power Plant Contractor shall configure correction equipment, or utilize large synchronous motor drives to achieve this overall figure.

Photocells should be used in the control of all external lighting circuits - building, area and road lighting, to reduce the energy used and improve the balance of natural and artificial light. Inside buildings, all circuits should be switched to allow unused facilities to be kept dark, whilst dimmer controls should be considered for all control rooms, meeting rooms and conference rooms.

7 BASIC DESIGN

The electrical system shall be designed to distribute safe and reliable power, utilizing new, state of the art equipment and materials. It shall be designed for continuous operation, providing adequate and reliable power supplies under steady state and motor starting conditions.

The electrical systems shall be designed and implemented to recognized and generally accepted good engineering practice, in accordance with International Engineering Standards,

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Kazakh Government Safety and Security Requirements, Industrial Standards and Codes and Manufacturer's Installation Requirements as appropriate.

All work related to any Kazakh Electricity Company transmission lines or equipment that will be owned or operated by said organization shall be in accordance with that organization's published standard requirements, unless specified otherwise. Any new transmission line link to the Integrated Petrochemical Facilities designed and installed as part of this Project, will be maintained and operated by Company.

7.1 Power distribution

A block diagram of the power distribution for the power-plant is as outlined in the Annexure-1.

The Power Plant will supply:

- Electrical power as detailed further below
- Primary steam (not exported) to drive the steam turbo-generatos
- High-pressure (HP) steam;

In preliminary analysis and the feasibility study, the Power Plant configuration comprises:

- Four (4) identical gas turbo-generators of ISO rating in the approximate range of 50 MW, each with its associated HRSG;*
- Two (2) condensing steam turbo-generator of approximately 55 MW with air-cooled exhaust condenser.

* Contractor is free to propose the different configuration as per the chosen size and ISO rating of the GTG.

Electrical power is primarily produced by electrical generators (10 or 11 kV nominal voltage level) driven by four (4) gas turbines and one (2) steam turbine. The Power Plant generators will be connected through step-up transformers 10 (11)/110 kV to the Main Electrical Substation nominal 110 kV bus bar. The IPC substations are powered through the number of 110 kV switchgears. The 110 kV substation is also connected to the 110 kV Intake Switchyard making possible of import/export power to the grid or wotking of the Power Plant in paralel mode.

The power plant loads should be distributed equally, or as close as possible, between the busbars of the switchboards at the main substations. The main feeders and switchgears at

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each of the distribution line shall be designed with the capacity to supply the total power for the initial and future known loads, with 20% spare capacity for unknown future expansion.

The electrical distribution network will be based on redundant feeders from the Power Plant or else from the main site substations to the unit substations. The battery limits are the downstream connection points of switchgears.

7.2 Voltage Levels

The following definition shall apply:

- LV (low voltage) : up to and including 1kV A.C. and 1.5kV D.C.
- MV (medium voltage): up to and including 72.5kV A.C.
- HV (high voltage) : over 72.5kV A.C.

The voltage levels likely to be seen or utilized on site in Kazakhstan are as follows.

- Primary Voltage Level : 110kV, 3Ø, 50Hz, 3wires, earthing according to Kazakhstan Grid
- Distribution, MV Voltage Level : 10kV, 3Ø, 50Hz, 3 wires, Resistance earthed
- Distribution, LV Voltage Level : 690V, 3Ø, 50Hz, 3 wires, directly earthed
- Distribution, LV Voltage Level : 400/230V, 3Ø, 50Hz, 4 wires, directly earthed

7.3 Motor Applications:

- Motors above 400kW : 10 kV, 3Ø, 50Hz,
- Motors from 50kW to 400kW : 690 V, 3Ø, 50Hz,
- Motors from 0kW to 50kW : 400 V, 3Ø, 50Hz,

7.4 Other Applications:

- Lighting : 230Vac, 1Ø, 50Hz
- Small Power Circuits and Sockets : 230Vac, 1Ø, 50Hz
- Heat Tracing Circuits : 230Vac, 1Ø, 50Hz
- Control Circuits – MV Motors : 110Vdc
- Control Circuits – LV Motors : 230Vac, 1Ø, 50Hz

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- Control Circuits – Instrumentation : 230Vac, 1Ø, 50Hz
- UPS Instrumentation Power : 230Vac, 1Ø, 50Hz, unearthed (to be confirmed later)
- DC Power – Electrical : 110Vdc, unearthed
- DC Power – Instrumentation : 24Vdc, unearthed

7.5 Supply Tolerances

- Voltage variation under normal conditions : $\pm 5\%$
- Frequency variation under normal conditions : $\pm 1\%$

7.6 Voltage Drops

- Power Feeder (substation to substation, switchboard to switchboard) : 1% of supply voltage
- Sub-Feeder : 5% of supply voltage
- Power - Running Motor : 5% of motor rated voltage
- Power - Motor Starting : 15% of motor rated voltage
- Power – Lighting : 5% of the lamp rated voltage

7.7 Power factor

The overall system power factor, inclusive of reactive power losses in transformers and other distribution system equipment, shall not be less than 0.90 lagging at rated design throughout of the Plant. Where necessary, power factor correction shall be achieved by varying the excitation of synchronous generators/ synchronous motors.

Separate power factor compensation will be provided at the 10kV bus in the process substations (if required)

7.9 Plant load

A schedule of the installed electrical loads, the maximum normal running plant load and the peak load, expressed in kilowatts and kilovars and based on the plant design capacity when operating under the site conditions specified, shall be prepared. This shall be completed and updated regularly throughout the design stage of the Project and shall form the basis for provision of the necessary electricity supply and distribution system capacity.

The total electrical loads shall be calculated as follows:

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7.9.1.1 Maximum normal running plant load = $x (\%) E + y (\%) F$.

7.9.1.2 Peak load = $x (\%) E + y (\%) F + z (\%) G$.

where: E = sum of all continuously operating loads.

F = sum of all intermittent loads.

G = sum of all stand-by loads.

x, y, z are diversity factors.

The values of the diversity factors x, y and z must take account of the individual drives or consumers which make up the continuous, intermittent and stand-by loads, respectively. Subject to the above considerations, the following default values could be used for initial load assessments, or if the diversity factors have not been finalized:

x = 100% (some diversity may be applied to non-process loads,

y = 30%

z = 10%

8 SYSTEM STUDIES

Power Plant Contractor shall provide a preliminary electrical distribution network study and report for all areas within his responsibility, to confirm to Company that the proposed voltages used, short circuit levels anticipated, and equipment capacities being included by Power Plant Contractor, are acceptable within the overall Project parameters, and practical within the site boundaries of the Integrated Complex as well as the Power Plant. The study shall be performed using industry recognized software (such as ETAP, EDSA) approved by Company, with all input parameters clearly defined, and all results interpreted for Company evaluation.

During the detailed design, this network study shall be expanded and enhanced to include all levels of distribution and all consumers on an individual basis. The final study and report is expected to include any power generation, packages, offsites and earthing systems within the Power Plant Contractor areas of responsibility, together with a clear definition of interfaces to other areas and systems outside Power Plant Contractor control. The final network model and software system shall be updated to "as built" status, and handed to Company for maintenance, operations and future expansion activities.

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8.9 Relay Coordination Studies

During the detailed design, the above studies and equipment selection will highlight protective devices, timers and other circuit elements which will interact to produce a graded and operable distribution system. Coordination studies for these devices shall be undertaken – perhaps as part of the above study software – to produce tables and data setting sheets which will be utilized as part of the set-up and pre-commissioning activities. These setting sheets will be updated to “as built” status, and handed to Company for operations and maintenance purposes. Load flow study shall confirm acceptable voltage drop for normal operation at full load from Generator to end consumers. Relay coordination shall indicate correct time cascade from upper relay to lower relay. Relay setting is in this scope.

9 SUBSTATIONS & SWITCHROOMS

Substations are anticipated to be elevated cable basement construction with outdoor transformers. Contractor shall consider suitable extreme weather condition protection for all outdoor and indoor equipment. The substation shall be a concrete building. Transformers, Switchgears, Bus Ducts, Batteries, charger shall be arranged according to a sound engineering practice and must meet or exceed IEC 61936-1 standard. Sufficient height shall be provided in cable-basement (minimum clearance from bottom of the beam to the floor)

Substation buildings shall preferably be located in non-hazardous areas, however a full Building Risk Assessment (BRA) shall be performed to determine if any substation structure needs to be classified as blast resistant. Should this classification be required, the mounting of equipment directly onto the substation walls shall be forbidden. All substations shall have fire rated external walls, doors and fixtures. The main rating is anticipated to be of 2 hours duration.

Each substation will have a transformer yard adjacent to one of its longer sides, complete with intermediate firewall separators for transformers, maintenance access space, and an outer fence to prevent intrusion by non-certified personnel.

A typical unit substation will include, but not be limited to, the following major equipment:

- MV/MV Transformers with dedicated Neutral Earthing Resistors
- MV/LV Transformers with Bus Ducts
- An elevated air-conditioned building containing electrical switchgear room, communication room and battery room

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- MV indoor GIS (Gas insulated switchgear)
- MV indoor double ended metal-clad switchgear
- MV indoor “intelligent” control gear boards (MCCs)
- LV indoor double ended metal-clad switchgear
- LV indoor “intelligent” control gear boards (MCCs)
- Rectifier/Chargers and batteries for the 110Vdc system
- Rectifier/Chargers, batteries and inverters for the electrical 230Vac system
- Power Monitoring & Control System equipment.
- Fault and Sequence of Event Recorder
- Digital Fault Recorder
- Dedicated FO patch panels for protection, PMCS, and “intelligent” control gear systems
- Fire Alarm Panel
- Building Services Panel (HVAC)

The substation shall be constructed above grade, but with an elevated cable basement below the main floor. The transformer yard will also be at grade, with the adjacent substation firewall extended down to grade level. All other sides of the cable basement may be open (protected by wire fencing), except on blast- resistant buildings. The substation will not have windows, but shall have at least one personnel door and one equipment door, preferably located at opposite ends of the building. The equipment door opening shall be big enough to accommodate the largest assembled unit of equipment, and shall use standard size doors with removable panels above, if required.

Underground cable feeders shall be installed from the power station and routed through the facilities to the substation transformer yard. Bus ducts from the transformers shall enter the substation through the adjoining wall, above the height of the installed switchgear. Power and control cabling to equipment and to process area consumers shall exit the substation through fire-rated cable transits located in the floor, and be supported on cable trays suspended in the elevated basement area. Internal cabling may be installed on cable trays above the panels, and, where necessary, exit the building through fire-rated cable transits in the wall, or in the cable basement.

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The substation layout shall provide space for the extension of switchgear and controlgear, over and above that specified elsewhere. It is anticipated that this space is sufficient for one additional tier of equipment per bus section.

A minimum vertical clearance of 450mm above equipment shall be provided, measured between the highest equipment section and the bottom of the lowest bus duct, roof beam, air conditioning duct or overhead cable support system. The building floor shall be level and smooth, and, where recommended by the Manufacturer, strips shall be provided to aid the withdrawal of switchgear carriages, installed level with the finished floor.

The interior of the building shall have a positive pressure HVAC system, to ensure that dust and unwanted gases are excluded from the interior.

Batteries will be located in a separate room on racks, with protection against acid spill and with redundant fan extraction system direct to the outside of the building. It is anticipated that eye-wash and/or safety shower facilities shall be provided in these rooms. All battery rooms shall be classified as a zone 1 gas group IIC to account for the presence of hydrogen. Therefore electrical equipment within the battery room such as light fittings and socket outlets shall be suitable for the area classification.

9.9 Telecommunications

Each substation building shall have a dedicated communications room for the marshalling of telephone and data communications. This shall include connection to the plant LAN and links between the substation, control room, process area and other subscribers required for voice and data transfer. Power shall be derived from the substation small power distribution board.

In addition to the above, dedicated networks that serve as communication links for the following systems shall be provided

- Between substation monitors and PMCS system server(s)
- Between substation protection networks
- Between substations and SRRs for “intelligent” controlgear

The equipment required for these networks shall be provided in the substation switchroom.

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9.10 Fixtures and Fittings in Substation

Contractor shall be responsible for fitting out the substations with the normal operating equipment required to provide a fully working environment for Company to take over at mechanical completion. It is anticipated that these fixtures and fittings shall include, but not be limited to, the following:

- Desks, tables and chairs for Operations Personnel
- Drawing racks and filing cabinets
- Storage cabinets
- Work tables or benches, including facilities for circuit breaker testing
- Special tool storage units
- Clock
- Notice boards and facilities to wall mount drawings
- First Aid Station

10 ITEMIZED EQUIPMENT & MATERIALS

10.9 Selection Criteria

All equipment and materials provided shall be of proven manufacture and well constructed. They shall also be highly reliable, of current - but established – design, and of recent manufacture. The design shall, where practically possible, be future proof, by incorporating the latest proven technologies. The equipment and materials shall be suitable for continuous operation. Design, engineering and manufacture of all equipment shall use good engineering practice, and shall make provision in the design for safety, ease of operation and maintenance. All equipment shall be of a fully proven type-tested design.

Materials, which are known to pose a risk to health, must be excluded as far as reasonably practicable. Where exclusion is not practical, Hazard Data Sheets shall be provided to Company with risk assessments and proposed risk management procedures.

Asbestos, or products containing asbestos, shall not be used. For each main equipment, from generators to switchgears, Contractor is expected to provide a technical bid evaluation.

10.10 Transformers

In the Power Plant the main POWER transformers are :

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- 10 (11)/110 kV rated step up transformers in the Power Plant (one per generator).

Other transformers will be specified by Contractor

Spare Capacity

All transformers ratings shall include a minimum of 20% spare capacity above the known load at start up. Contractor shall submit transformer sizing calculation. Transformers shall be ONAN and comply with IEC standards.

All transformers, Distribution or Power, shall be connected to a double ended switchgear,. Each one shall be rated for the total requirement to ensure a 100% +20% spare redundancy when the other transformer fails. The self cooled rating of transformers shall be calculated taking into account the design temperature rise, de-rated if necessary, for continuous operation at high ambient temperatures.

Power transformers shall be installed outdoor. Wherever possible, they shall be shielded from any sources of high-pressure shock waves that could be generated from vapour cloud explosion within the process areas. When required, they shall be protected against direct sun rays.

Power transformers shall be isolated from each other by firewalls, and have drainage facilities around their concrete pads - preferably covered with suitable aggregate fill. Transformers shall have either an individual or a common containment pit for oil spills, sized for the oil quantity in the largest transformer.

Oil containment pits and clearances in air shall comply with known standards such as: IEC 61936-1. For further details refer to specification 108184-0000-EL-SPC-0010 (Power Transformer)

Distribution transformers

In this project, there are three types of distribution transformers:

- Distribution transformer 10/0.4kV shall be dry type, rated 2500kVA maximum, (air natural) with cast resin or epoxy encapsulated coils. They shall be installed indoor.
- Distribution transformer, 10/0.69kV, with rated power up to and equal to 2500kVA, shall be dry type, (air natural) with cast resin or epoxy encapsulated coils. They shall be installed indoor.
- Exceptionally transformer, 10/0.69kV, above 3150kVA, may be used if no possibility to split into smaller pieces. They shall be oil-filled conservator type(for high humidity

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locations - with a membrane, in order to prevent “direct oil-to-air” contact). They shall be installed outdoor. They shall be isolated from each other by firewalls, and have drainage facilities around their concrete pads preferably covered with suitable aggregate fill. Transformers shall have either an individual or a common containment pit for oil spills, sized for the oil quantity in the largest transformer. Oil containment pits and clearances in air shall comply with known standards such as: IEC 61936-1.

All distribution transformers shall have externally operated, off-circuit, primary tap changers, with full capacity at any tap: middle tap and taps at ± 2.5 and ± 5.0 %.

Transformer primary windings, MV range, shall be configured for incoming cables uprising from below. They shall have side wall bushings with right angle plug in connectors, IP56. Cable holders shall be welded to the tank. They shall be strong and far enough not transfer heat of the tank to the cable.

Transformer secondary windings shall be configured for top or side entry according to specific project requirement. If large current is involved, LV sealed bus ducts shall be used (currents more than 1600A)

Neutral of transformers

For power transformers, the neutral circuit shall be uniform insulation, from neutral to bushing. All power transformers shall have neutral resistors.

10.11 HV and MV Switchgears

The switchgear construction shall be :

- Gas insulated switchgear will be provided for main power distribution at 110 kV. GIS will be double-busbar system in compliance with IEC standards
- Metal clad, indoor type, with withdrawable type in compliance with IEC standards at 10kV for powerplant and process units. Vacuum/SF6 circuitbreakers shall be used for feeders/transformers and for motors fuse-combination vacuum contactor starter units will be used.

Protection relays shall be microprocessor based, programmable, and communication type units. Provisions for an automatic transfer system shall be made. Voltage and current transformers of appropriate ratio, output class and accuracy for protection and metering shall be provided.

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All MV switchgear shall be extendable from both ends. Earthing switches shall be provided to earth the incoming and outgoing circuits. Suitable mechanical key interlocks shall be provided to ensure that circuits are isolated before earthing switches can be closed and earthing switches are open before energizing the circuits.

All circuit breakers in the switchgear assembly shall be installed in single tier formation. 10kV motor feeder units may be installed in double tier formation.

At least one motor and one transformer feeder cubicle (fully equipped) shall be provided as spare on each section.

10.12 Low voltage Switchgears

There are two levels of low voltage switchboards:

- 690V for motors from 50 kW to 400kW
- 400V for general use and motor from 0 to 50 kW

LV switchboards are made of:

- Two incoming circuit breakers
- One bus tie breaker
- One Automatic Transfer System
- Outgoing breakers arranged in drawers

Ratings of breakers shall be figured out later by Contractor when consumer list become available. Consumer list is in the scope of Contractor. Current limiting circuit breakers and HRC fuses are permitted. Control diagrams shall be given. Applicable standards are IEC.

Switchgear and Controlgear assemblies shall be extendable at both ends. The switchgear and Controlgear shall be accessible from the front and suitable for back to back installation.

10% fully equipped outgoing circuit compartments shall be provided. Protection relays for low voltage motors, feeders and main incomer and bus sections shall be microprocessor based programmable type with communication facilities to the PDCS system.

10.13 LV Bus Ducts.

At the low side of large distribution transformers (with secondary current higher than 1600A), multi-cables in parallel per phase should be avoided where possible because of un-even

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heating between phase. In this case, metal enclosed Busducts, IP 56, shall be used to connect the secondary terminals of step-down transformers to LV switchgears. For small size transformers, underground cables are largely used.

Bus duct shall be adequately supported and use proper expansion joints. End connections for the bus ducts shall be proven products. They shall not be site fabricated units.

11 SCADA SYSTEMS

Monitoring of electrical systems shall be implemented throughout the facilities. This project will install only equipment and systems suitable for connection to a supervisory system. Power Plant Contractor shall ensure that all equipment provided can communicate with the central system, and that the vast majority of the modules and system are from the same Manufacturer.

11.9 Power Management System

It is envisaged that there will be a Power Management System (PMS) developed for the Power Plant. Power Plant Contractor shall also provide a Power Distribution & Control System (PDCS) to operate the IPC electrical system.

The site-wide PDCS shall be located in the Main Control Building, together with a local client PC. They shall communicate with monitors in the various substations via a dedicated Ethernet network. The network between buildings will be fibre optic based, with an industrial Ethernet switch and a wire patch panel at each location.

The PDCS shall be used for:

- Monitoring and controlling all substation power transformers, 110 kV switchgears, 10kV switchgears
- Monitoring all 690 V and 400V switchgear incomers and bus tie circuit breakers
- Monitoring and controlling all 10kV outgoing feeders
- Substation individual and group alarms
- Substation environmental data and HVAC status

Power Plant Contractor shall hard wire status information to appropriate modules in the substation, using I/O extenders where necessary. A detailed points list shall be developed, and updated as necessary during the detailed engineering.

The PMS shall consist of three general components:

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- “Intelligent” Meters or Protective Devices to take measurements, generate data and control actions
- Fibre optic links to communicate data with Central Control Building PMCS computer workstation
- Communication links to enable export of data to a local portable computer. The portable computer shall not form part of the Power Plant Contractor’s scope of work.

The PMS communication network consists of Passive Hubs and Active Hubs. Each shipping section of switchgear shall have a Passive Hub. All PMS modules within a shipping section shall be connected to the Passive Hub in that section. Each Passive Hub shall be connected to one of the Active Hubs within the substation. Signals from the Active Hubs shall be grouped and transmitted around the network. Monitors will be provided in each substation to locally monitor all connected electrical equipment and the network status.

Power Plant Contractor shall ensure that the PMS software is energy conservation oriented. This network will interface, through suitable firewalls, with the DCS/PMS to pass data required by the Process Operators for monitoring and control of the process and utilities areas.

The system, including many of the remote monitors, must be provided by a single Manufacturer. Power Plant Contractor shall provide a Subcontractor Commissioning Engineer to supervise the installation, pre-commissioning and commissioning of the elements of the PMCS system.

11.10 Alarms

All major equipment and all packages having an instrument, control or electrical content, together with all electrical distribution network equipment, are expected to generate status and alarm signals (minimum common alarm) to allow their operation to be remotely monitored. Signals reflecting the electrical status of the equipment should be connected back to the substation from which their power is derived. In the Power Plant all alarms must be reported: electrical, mechanical, steam, firing, utilities.

11.11 Control and Monitoring

The Power Management System (PMS) will be linked to the individual Gas Turbine (GTG) and Steam Turbine (STG) generator control systems to give high level control of the power generation, power flows in the network, synchronization with the Kazakh Grid, and between

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generation sources. This PMS system will require inputs from all the IPC Facility substations to measure and predict process, utilities and infrastructure loads. Power Plant Contractor shall make provision within his areas of responsibility to make such signals available to the PMS. The extent of this PMS includes the 110kV grid connection and the residential village.

In addition to the PMS at the IPC, there should be a Load Shedding System (LSS) developed to divest the distribution system of blocks of electrical load, in the event of unpredicted or catastrophic loss of generator or grid supply. This LSS for the Power Station will also be interfaced to the IPC Facility substations in the form of trip signals for the various load circuit breakers and motor controllers requiring to be disconnected from the network, and feedback signals on their status. Power Plant Contractor shall make provision for all required control and status signals.

For fast signals (control), hard wire shall be used. Optical fibre is accepted for slow signals (status). It is also envisaged that a Power Monitoring & Control System (PMCS) will be provided around the IPC substations to allow Operators access from a central control point to interrogate and check performance of the distribution network. The PMCS will be based on intelligent modules and protection devices, located in substation switchgear and controlgear, linked together on a dedicated Ethernet/fibre optic network and driven by a system server from the main Control Building. This network will interface with the DCS/PMS to pass data required by the Process Operators for monitoring and control of the process and utilities areas.

All controlgear for the process and utilities will be under selectable control of the DCS/PMS. The interfaces between systems will be provided in the form of cabinets for DCS/PMS and ESD signals located in the substations. Power Plant Contractor should consider “intelligent” controlgear as an option, allowing the DCS/PMS interface to be a communications based fibre optic link, rather than hard-wired.

12 INSTRUMENTATION POWER SUPPLIES

Dual redundant, 230V, single-phase UPS systems shall be provided to supply instrumentation loads. The systems shall be dedicated to the PCS/DCS system, with 30 minutes battery autonomy time at full rated load, and the Fire & Gas system, with 24 hours battery autonomy time at full load. The connected loads will consist of, but not be limited to, field instrumentation, instrument control systems and instrument power supplies. The UPS shall have a minimum of 20% overall spare capacity for instrument system development. Load spare capacities shall not be cumulative. Each UPS system shall include

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- Dual Rectifier/Charger Units, each rated at 100% of the load + battery charging
- Dual Batteries, each rated at 50% of the load for the full autonomy time
- Battery Isolation Circuit Breakers (switches)
- Dual Inverter Units, each rated at 100% of the load
- Static Transfer Switches
- Maintenance By-Pass Switch
- Bypass Transformer
- Main Distribution Circuit Breakers, to feed the Instrument Distribution Panels
- A suite of matching panels, located in an electrical room of the SRR or Main Control Building, to house the main system components

Batteries shall be located in a separate room on racks, with protection against acid spill and with redundant fan extraction system direct to the outside of the building. It is anticipated that eye-wash and/or safety shower facilities shall be provided in these rooms. Each room shall have a dedicated door to the outside area.

Batteries shall be recharged by the system to 90% of their capacity within 8 hours. Battery life under normal operation, and with the above recharge cycle, shall be guaranteed to be not less than 15 years.

The rectifier/chargers shall be fed from separate MCC (SRR), or separate LV bus sections. If emergency power is available in the source substation, one of the rectifier/charger units shall be fed from this supply. The bypass supply may be derived from any of the above sources.

It is anticipated that a web based monitoring system for the UPS shall be supplied and connected to the site PMCS via a communications hub.

13 DC POWER SUPPLIES

In the substation, one 110 Vdc battery system shall be provided for the power requirements of switchgear control and protection equipment and miscellaneous items such as substation annunciators. The systems shall be dedicated to the equipment within the substation, with 4

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hours battery autonomy time at full load. The system shall have a minimum of 20% overall spare capacity for substation development.

Each DC system shall include

- Dual battery chargers, each rated at 100% of the load + battery charging
- Dual batteries, each rated at 50% of the load for the full autonomy time
- Battery Isolation Circuit Breakers (switches)
- Fully rated blocking diodes
- DC panel board, with circuit breakers for bus closing supplies, bus tripping supplies, annunciator power, etc.
- Battery monitoring panels

Battery chargers shall have load-sharing capability.

Batteries shall be located in a separate room on racks, with protection against acid spill and with redundant fan extraction system direct to the outside of the building. It is anticipated that eye-wash and/or safety shower facilities shall be provided in these rooms.

Batteries shall be recharged by the system to 90% of their capacity within 8 hours. Battery life under normal operation, and with the above recharge cycle, shall be guaranteed to be not less than 15 years. The battery chargers shall be fed from separate MCC, or separate LV bus sections. If emergency power is available in the source substation, one of the charger units shall be fed from this supply.

It is anticipated that a web based monitoring system for the DC power supply shall be supplied and connected to the site PMCS via a communications hub.

14 MOTORS

Motors will be as per Vendor specification (Induction Motors)

15 VARIABLE SPEED DRIVES

LV VSDs are foreseen for cooler/fin fan trim duties, and possibly for some pump controls. It is proposed not to install these in MCCs, unless they are small single units, but to mount them in

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groups within “wardrobe” type panels, having a common bus and feed from an LV switchboard circuit breaker.

Should any HV or MV drives be required, they shall be supplied as separate, stand alone units located within the substation. Contractor shall study the effects of large cooling systems for such drives, including required air flows, water systems, and noise levels inside the substations.

The panels shall be distributed logically across the LV switchboard split bus, giving cooling availability under all supply conditions. It is not required for the individual VSD to have fixed speed bypass facilities, however it shall be possible to take any single VSD out of service, without affecting the performance of the other units. Panel cooling shall be provided, either in the form of redundant central cooling fans, or utilizing the individual VSD cooling fans – as long as these can be shown to share airflow correctly.

Control of the drives shall be generally similar to that of the standard MCC arrangement, with hardwired 4-20 mA speed control/feedback signals to the PCS/DCS via the interface panels. If “intelligent” MCCs are selected, Contractor shall consider providing the control and feedback via a communications bus system. Contractor shall demonstrate to Company that the VSD chosen, and their mounting arrangements, does not have an adverse effect on the overall harmonic content of the power supply, or the electromagnetic stability of the substations.

16 LOCAL CONTROL STATIONS

Generally, process motors shall be automatically controlled and each motor shall have a local field control station comprising of a START, stay-put lock-off STOP, MANUAL/OFF/AUTO selector switch and start/stop indicator lamps adjacent to the motor. The method of control to be provided shall be as defined on the P&ID drawings. Ammeters showing motor current will be provided in the local control stations for MV motors. Ammeters will be operated from 1A secondary mounted in the MCC

For motors that are automatically controlled from the PCS/DCS, signals shall be communicated between “intelligent” control gear in the substation and a communications

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interface in the SRR, through patch panels and fibre optic cables. Controls shall include, but not be limited to:

- Motor start
- Motor stop
- Motor status (running)
- Motor drive fault
- Motor in auto selection (available to run)

Motors which do not have the local control station adjacent to the motor (such as elevated fan motors, etc) shall, in addition, have a hard wired EMERGENCY STOP button provided adjacent to the drive, connected direct to the motor control circuit.

Control stations shall normally be installed on suitable steel supports close to the motor they control and shall be wired directly to the motor starter in the Controlgear.

17 LIGHTING & SMALL POWER

A 400/230Vac, 3P+N+PE, lighting and small power distribution system shall be provided for substation, rack room, and other major buildings around the sites. In addition, outdoor panels should be located around the facilities, to provide local feeders in remote areas. All lighting and small power circuits shall be protected by suitably rated circuit breakers.

Light fittings shall be supplied from lighting distribution boards containing MCBs to protect each sub-circuit and enable individual circuit isolation. Light fittings shall be rated to operate from a 230V single phase supply. Lighting distribution boards shall be three phase 4 wire and where possible shall be located within a safe area, preferably within buildings.

Use of high pressure mercury vapour lamps shall be maximized as far as possible for illumination at grade and operating platforms, below pipe racks, etc. throughout the plant. However, light fittings in strategic locations and for emergency lighting shall be fluorescent type.

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Industrial/decorative fluorescent lighting fixtures with electronic ballasts shall be used for illumination in substations, control rooms, offices, etc.

Lighting fittings for general illumination shall be located as close as possible to instruments, gauges etc. as to avoid special lighting for these devices.

All outdoor plant lighting, street lighting, and flood lighting shall be controlled by photo electric cells. Site wide, the use of local switches shall be kept to a minimum. For all street lighting and flood lighting high pressure sodium vapour lamps shall be used.

Lighting/receptacle distribution Panels shall be located in the area being served. All outgoing circuits in Panels located in plant area shall be controlled by double pole miniature circuit breakers. At least 20% spare outlets shall be provided in each lighting and receptacle distribution Panel.

Lighting fittings specifically installed for gauge glass illumination shall be on separate sub-circuits and shall be controlled by a locally mounted switch.

Aircraft warning lights shall be installed on all structures 45 meters and above and in accordance with ICAO Publication Annex 14 and local standards. Lamps used for aircraft warning lights shall be of long life type (minimum 20000 hours)

All bulk material for lighting/socket system will be made of one of the following material:

- GRP (Glass reinforced polyester)
- Hot dip galvanized steel
- Copper free aluminium alloy for hazardous areas only (if any)

All Lighting and small power equipments which are installed outdoor shall be suitable for Ex-d IP65 degree of protection.

All lighting fixtures and power outlets shall be suitably certified for the area in which they are installed (normally EEx d/de), and be provided with the appropriate documentation.

The small power scheme shall be designed to ensure power is readily available for the use of mobile or hand held equipment around the facilities. Single phase power shall be distributed

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around the buildings and the process areas, whilst three phase power distribution will be distributed around the process and external areas only.

All receptacles shall be standardized, to ensure that only one type of plug per system is required, and can be used with all such receptacles installed within the plant. The internal convenience plugs shall be normal ones available in the Kazakh market, and Contractor shall provide a suitable number for initial Company use. The external convenience and welding plugs should be IEC standard equipment type, and Contractor shall provide a reasonable number of each type for Company use.

Receptacles which are to be installed in prospective wet locations shall be protected by circuit breaker, having ground fault circuit interruption facilities.

Welding socket outlets and convenience socket outlets will be provided in the plant. Illumination levels and further details will be as per spec. 108184-0000-EL-SPC-0023.

18 CABLING SYSTEMS

Contractor shall develop a cable specification and shall prepare a cable sizing calculation for 110 kV, 10kV and LV power cables. Following aspects shall be considered in cables sizing:

- Thermal short circuit capacity.
- Voltage drop.
- Continuous Current rating.

All feeder and sub-feeder cables shall be sized for continuous duty. All cable sizes shall take into account the ambient/soil temperature, soil thermal resistivity and group de-rating factors.

Cables shall be sized to withstand the stresses due to short circuit current equivalent to the rating of switchgear and the fault clearing time of protective devices. All cables will be supplied and installed in single lengths, unless the length exceeds the standard drum-length for that size.

18.9 110 kV cables

- XLPE insulated, armoured

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- Single core, Copper : 300 mm² size, or according to cable sizing calculation
- Applicable standards are IEC-60332-3, part A

18.10 10 kV cables

- XLPE insulated, armoured
- Single-core, Copper for feeders : according to cable sizing calculation
- Multi-core, Copper for motors : according to cable sizing calculation
- Applicable standards are IEC-60332-3, part A

18.11 LV Cables (690V and 400V)

- XLPE insulated, unarmoured
- Multi-core or single core : minimum size 2.5mm² for power and 1.5mm² for control)
- Applicable standards are IEC-60332-3, part A
- Applicable standards are IEC-60331 for critical services cables, Fire and gas, UPS outgoing cables In general cable laying will be:
- From Generator to Step-up transformer: underground (directly buried).
- From Step-up transformer to 110 kV GIS incoming breaker: underground (directly buried).
- From 110 kV GIS to the 110/10kV step-down transformers: underground (directly buried).
- From 110/10kV step-down transformers to process units : underground (directly buried by others)
- 10kV cables in power plant for feeders : underground (directly buried)
- 10kV cables in power plant for motors : underground (directly buried)
- Low voltage cables : aboveground as far as possible in metallic cable-trays

All HV cables in directly buried trenches will be laid in a single layer. Cables shall be laid on 75 mm bed of clean compacted sand and covered by 75 mm of clean sand with cable protective tiles laid on top. The trench shall then be backfilled to grade level. Cable protective tiles are not

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required in paved areas of the plant. Where cables are laid in two layer formation, sand filling between the layers shall be 150 mm.

All motor control cables shall be laid alongside their respective power cable.

Whenever cables emerge from underground to connect to equipment they shall be sleeved in U.V. resistant pipe to at least 150 mm above finished grade level. Pipes shall be sealed after cable installation, with weatherproof compound.

Cables crossing roads in general shall be installed in concrete encased PE pipes and fitted with draw ropes. At least 20% spare pipes subject to a minimum of two, sealed at both ends, shall be installed for future requirements.

Cable tray or ladder racks supported from structures shall be generally used for overhead multiple cable runs. Cable trays shall be ladder type with rungs, 100mm high, with standard widths 300mm, 450mm, 600mm, 750mm and 900mm. Individual cables may be clipped and supported directly from structure. All cable trays/ladders shall be laid in horizontal formation supported at a distance of not more than 3m.

Cable laying on cable tray will be single layer with spacing for MV cables, single layer touching for LV power cables and double layer for control cables.

Cable tray covers shall be provided for mechanical protection (on the topmost layer in horizontal and vertical runs. Cable trays/ladders and covers, for power and control, shall be heavy duty, hot dip galvanized after fabrication.

Rungs for the cable ladders shall be welded to the side rails. Thickness of steel used for fabrication of cable tray/ladder/covers shall be 2 mm minimum. All cable ties shall be stainless steel plastic coated type.

All cables entering any piece of electrical equipment or fitting shall be glanded using brass double compression cable glands. All cable glands shall have an earth tag and PVC protective shroud. All cable glands to be used in Zone 1 and Zone 2 areas shall be Ex(d) (if any).

Cable numbers shall be marked on the cables along their routes and at both termination points. For underground cables the spacing between cable numbers along the route shall not

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exceed 5 m and for aboveground cabling 25 m. Cables shall also be numbered where they branch off from main route.

Underground cable markers shall be of stainless steel sheet on which the cable number has been printed by means of letter punches. For aboveground cabling, metallic/UV resistant plastic markers resistant to the atmospheric conditions shall be specified. All cable markers shall be tied to the cable with plastic coated stainless steel cable ties.

For underground cabling in unpaved areas, aboveground cable route markers shall be provided at every 25 m and at every change of direction in the routing and at both sides of road or pipeline crossing.

Adequate space (minimum 600mm) will be maintained between electrical power and instrumentation signal cables to avoid interference.

All field equipment with threaded entries shall be suitable for METRIC type threading. For unthreaded entries in electrical equipment, 3mm thick galvanized steel cable gland plate shall be provided. In case of single-core cables, the gland plate shall be of non-magnetic material.

19 EARTHING

Contractor shall provide a common earthing system which gives:

- Safety of personnel
- Protection against static electricity
- Protection against lightning discharge
- Protection against electromagnetic interference.

Ground networks shall be installed around substations, process units, structures, and other electrical installations. All of the ground networks shall be connected at two or more points to form an overall network. Equipment to be grounded includes tanks, vessels, skids, pipe racks, motors and other electrical equipment. All exposed conductive parts of all electrical equipment shall be bonded to the ground network. If there is no network nearby, they shall be connected to one or more dedicated ground rods.

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Contractor shall ensure co-ordination between the designs for both the earthing system, and the cathodic protection system.

Main earth grid conductor shall be 70 mm² bare copper conductor, except in substations which shall be 25mmx3mm bare copper flats.

The earth grid shall extend throughout the installation in the form of a main earth ring with branch interconnections to equipment and structures to be earthed. The metallic enclosures of electrical equipment shall be bonded to the plant earth ring. The metallic enclosures of nonelectrical equipment, e.g. vessels, shall also be bonded to the plant earth ring.

Earth grid interconnections belowground shall be carried out with thermal welds. Aboveground earth grid connections shall be carried out with compression connectors and earthing connection to equipment/structures shall be made by bolted connections.

Lightning and static earthing protection shall be provided by connecting steel structures, towers, vessels, tanks, etc., directly to the earthing system. Tall steel structures such as towers or structure columns, provided they are electrically continuous, shall be considered inherently protected against lightning by their connection to the earthing system. Bonds across joints may be used to ensure electrical continuity if necessary.

Electrically continuous structural columns may be used as down conductors by means of which elevated tanks, vessels, etc., shall be connected to the earthing system. Separate lightning protection systems shall be provided for tall masonry or concrete structures based on the concept of acceptable risk. The system shall be electrically and mechanically bonded directly to the general earthing system.

Flanges of metallic pipelines that have insulated linings shall be bonded to ensure electrical continuity. Flanged joints in un-lined or continuously welded pipelines shall be considered inherently bonded.

Pipelines shall only be connected to the earthing system where they enter and leave the battery limits or when entering or leaving hazardous areas.

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20 HEAT TRACING

Because of the severe weather conditions during the winter months, and the lack of spare steam capacity around the sites, it is anticipated that an extensive network of electrical heat tracing circuits will be required.

The PFDs and P&IDs will identify process lines, tanks, pumps, etc. requiring tracing, of which, some may be in classified areas. Local, dedicated, 3P+N, distribution boards in “safe” areas are foreseen, providing power to self-regulating tape circuits and thermostatically controlled lines.

It is anticipated that Contractor will derive all circuit data from the process data, piping isometrics, insulation details, and site information document, and produce the heating scheme from this base data. As an alternative, if early plant modelling is available, Subcontractor software packages may be used to produce the scheme. All such software is subject to Company approval before use on the project.

Individual circuits shall have adequate circuit breaker protection, and be installed using proprietary connection items, splicing and end termination kits, pipe straps and tape, and be fully identified using permanent labels for all lines and equipment. Manufacturer’s published installation details shall be followed at all times, and full “as built” documentation produced for the system.

The system shall be fully tested prior to the application of the insulation on the lines requiring protection, and again after insulation is completed to verify circuit continuity has not been compromised.

Monitoring systems shall generally be electronic, with status signals and data made available to the PMCS system through the substation hubs.

21 HVAC POWER SUPPLIES

HVAC systems in buildings will require single and three phase power to operate their equipment. Contractor may integrate these elements within his MCCs and distribution equipment, taking full regard of the special control and monitoring required by the HVAC

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system controllers. Alternatively, Contractor shall provide dedicated HVAC panels to distribute power to the elements of the system. These panels will be required to have full redundancy, as per the rest of the distribution system, but shall interface only with the HVAC controllers and modules, and not with the PMCS.

Contractor shall clearly identify where emergency power is required by the HVAC system, and ensure that this is available from the nearest substation.

22 E&I INTERFACES

A typical unit substation will include, but not be limited to, the following interface facilities:

- ESD Interface Panels
- Communications hubs for “intelligent” Control gear interfaces
- E/I Interface Panels, if “intelligent” Control gear is not provided

The ESD interconnecting system shall be hard-wired for all related status and permissive signals. These signals, from the various switchgear and control gear, shall be marshalled, and all necessary interface relays provided, in the interface panel.

23 PAINTING

Painting will in general be as per Vendor Standard for Protective Paintings and Application Requirements.

Major electrical equipment is anticipated to have the following colour scheme, which shall be extended across all sites and all facilities:

- Electric Motors/generators : Blue, RAL 5010
- Fire Water Pumps & Panels : Red, RAL 2002
- Transformers : Light Grey, RAL 7035
- Switchgear : Light Grey, RAL 7035
- Control gear : Light Grey, RAL 7035
- Utility Panels : Light Grey, RAL 7035
- Cable trays : Hot dip galvanized

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- Push-buttons : Green for START and Red for STOP
- Indicating lamps : Red (Energized circuit or running motor), Green (De-energized circuit or standstill motor), Amber (Alarm due to fault), White (Control Voltage available), White (Position signaling of a mechanical device), Green (Operating consent)

Electrical bulk material : Black (GRP) or Light grey RAL 9006 (Aluminium alloy) or Hot-dip galvanized finish

The protective finish provided on equipment should reflect the final location where the equipment shall be used. Outdoor equipment will be subject to an industrial petrochemical environment, with saline air and extensive windblown dust. Final equipment finishes shall be submitted for Company approval. Internal equipment is likely to be in HVAC controlled environments, and Subcontractor's standard finish may be acceptable for these locations.

24 PACKING, SHIPPING AND STORAGE

Packing, shipping and Storage will be as per the specification 01000-PU-SPC-0002 Project Packing, Marking and Shipping Instructions.

Equipment manufacturers/vendors will prepare specific instructions for the preservation and maintenance of equipment while stored at the construction site, from receipt at site to operational start-up. Such instructions shall include as a minimum the preservation and maintenance schedule, preservative materials, lubricants to be used etc.

25 NUMBERING SYSTEMS

Electrical systems shall include all power, earthing, lighting, small power (sockets, etc.) and trace heating systems and raceways. Equipment and cables shall be clearly and permanently marked as per the Project standards.

25.9 Tagging and Labelling

Power Plant Contractor shall confirm with Company all the tag numbers of permanent electrical equipment to be installed on the Integrated Project.

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